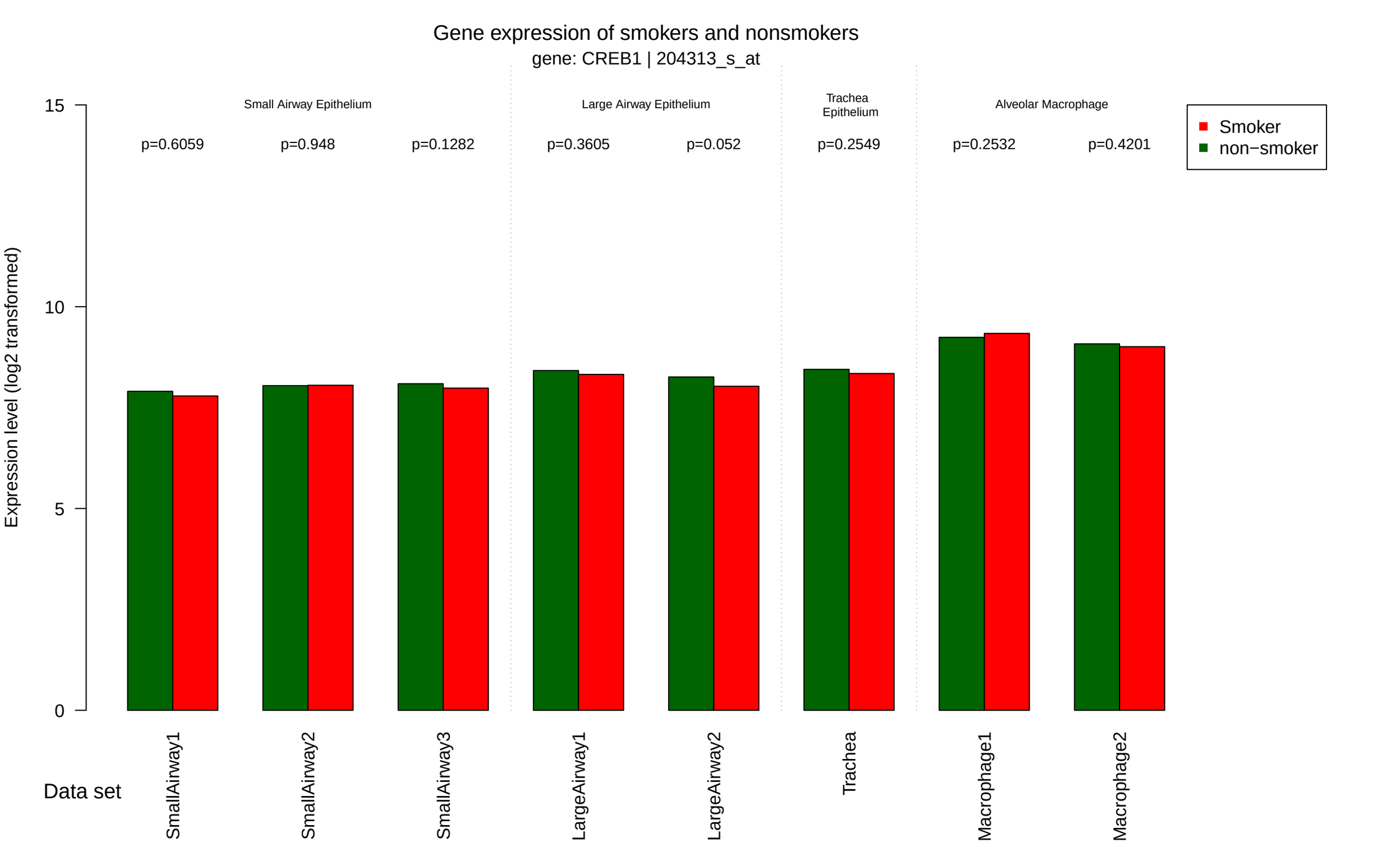
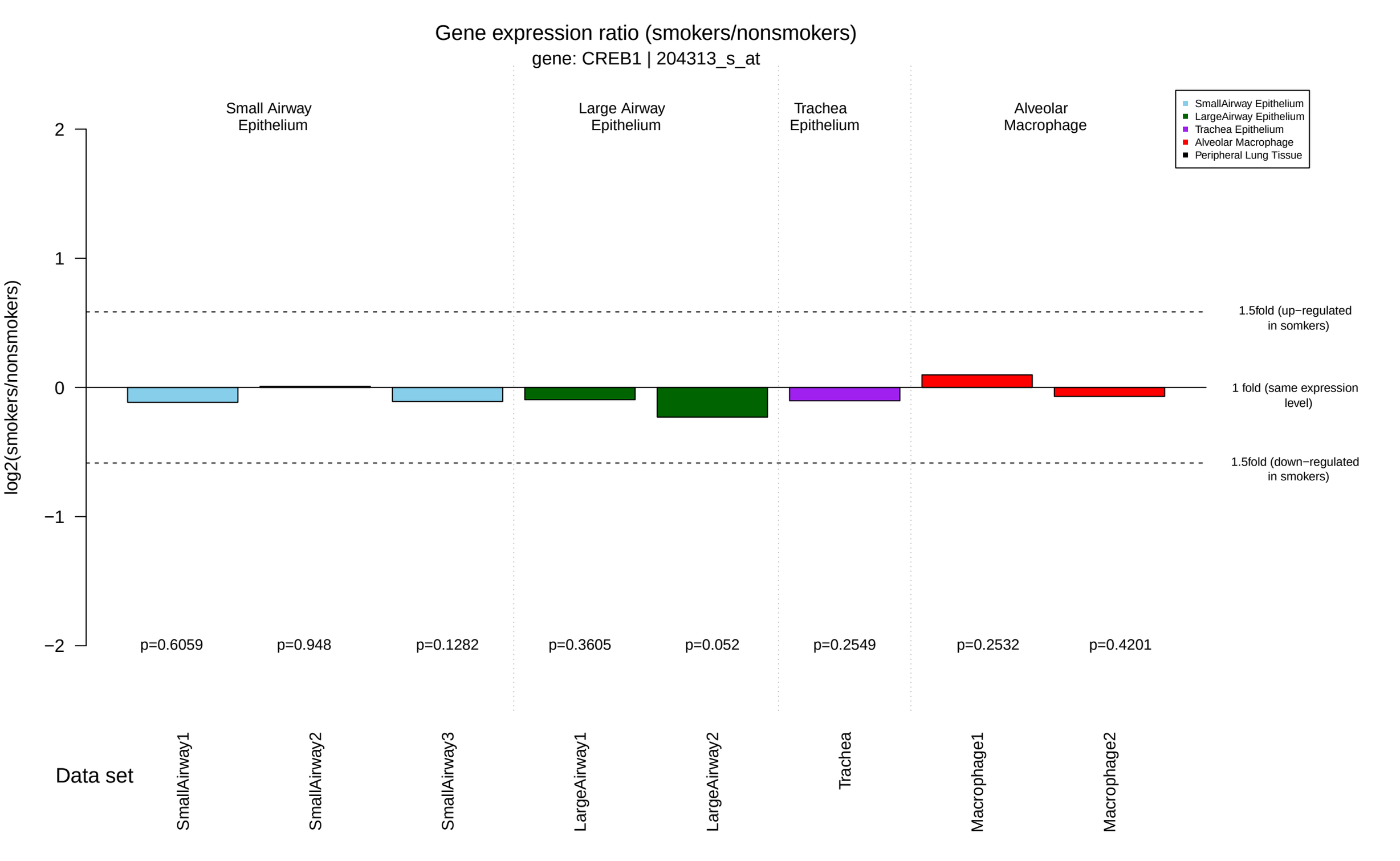
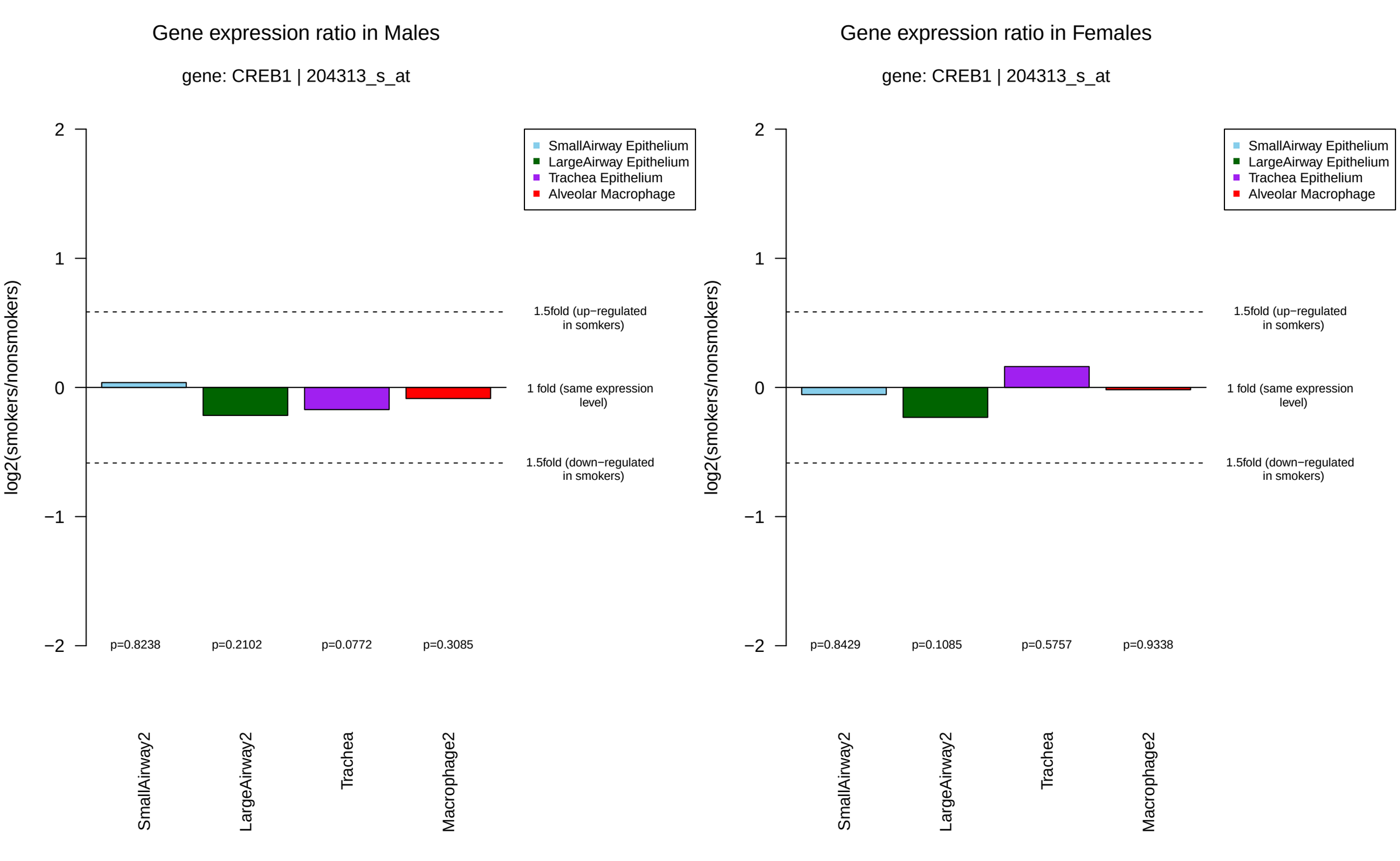
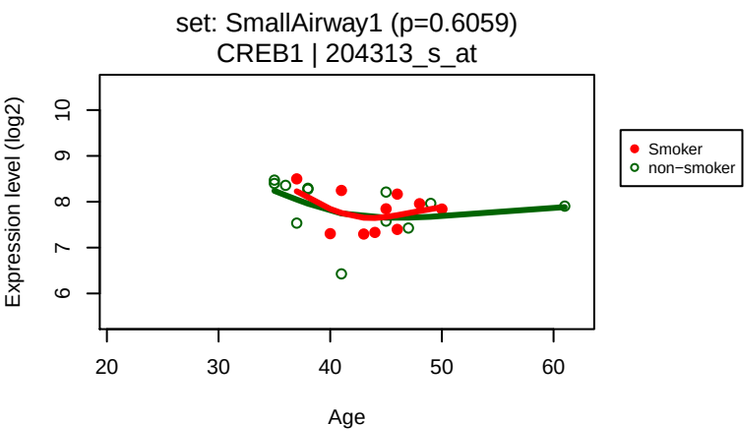
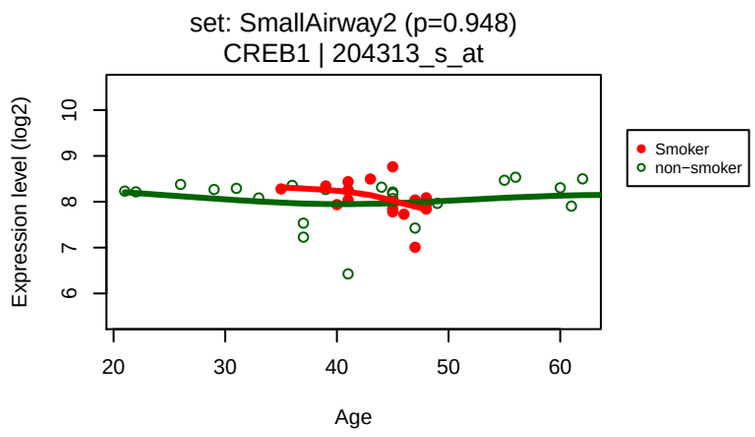
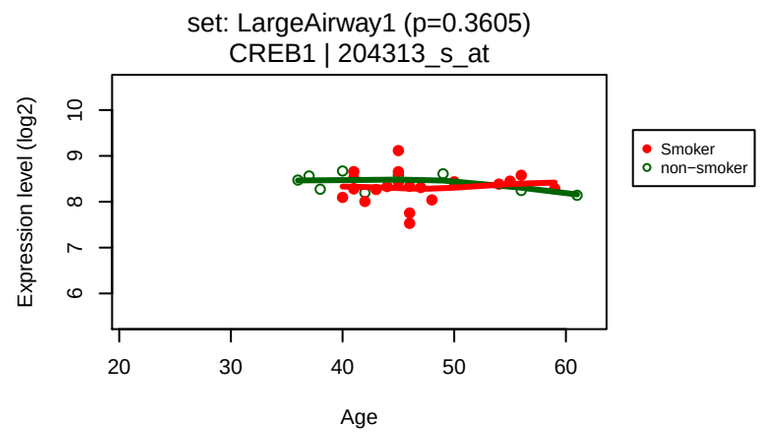

Gene expression of smokers and nonsmokers
gene: CREB1 | 204313_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6059
p=0.948
p=0.1282
p=0.3605
p=0.052
p=0.2549
p=0.2532
p=0.4201
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

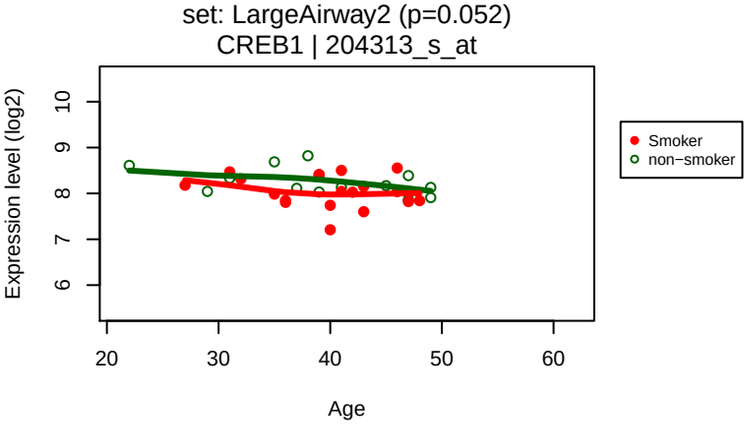

Gene expression ratio (smokers/nonsmokers)
gene: CREB1 | 204313_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6059
p=0.948
p=0.1282
p=0.3605
p=0.052
p=0.2549
p=0.2532
p=0.4201
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

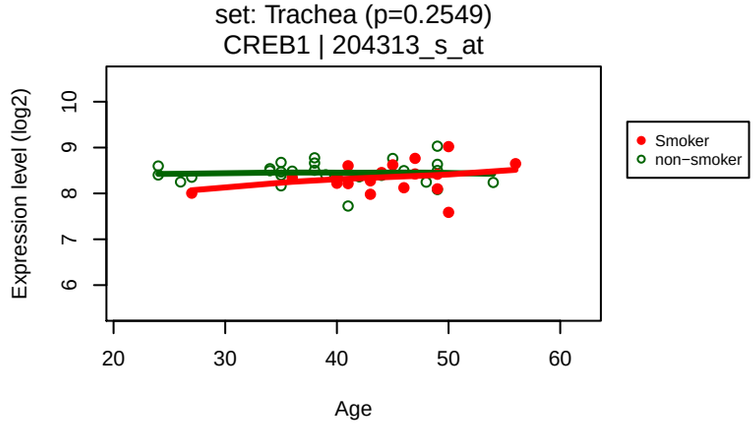

Gene expression ratio in Males
gene: CREB1 | 204313_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8238
p=0.2102
p=0.0772
p=0.3085
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CREB1 | 204313_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8429
p=0.1085
p=0.5757
p=0.9338
SmallAirway2
LargeAirway2
Trachea
Macrophage2

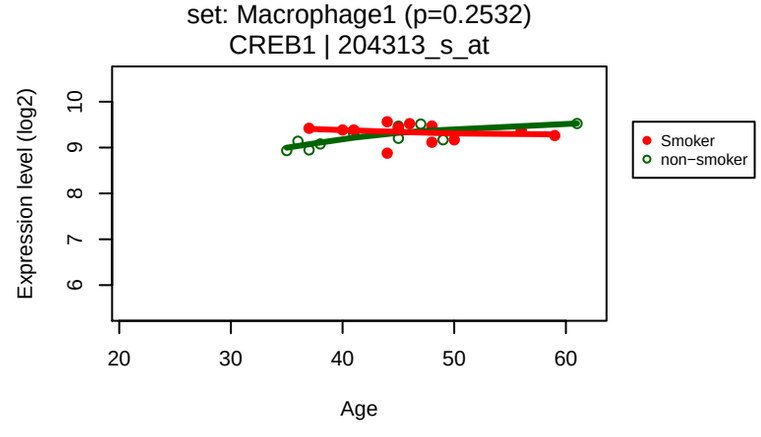

set: SmallAirway1 (p=0.6059)
CREB1 | 204313_s_at
Expression level (log2)
Age
Smoker
non−smoker

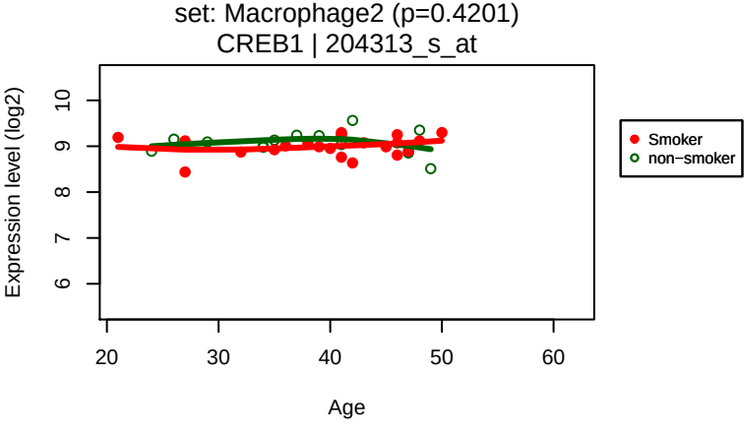

set: SmallAirway2 (p=0.948)
CREB1 | 204313_s_at
Expression level (log2)
Age
Smoker
non−smoker

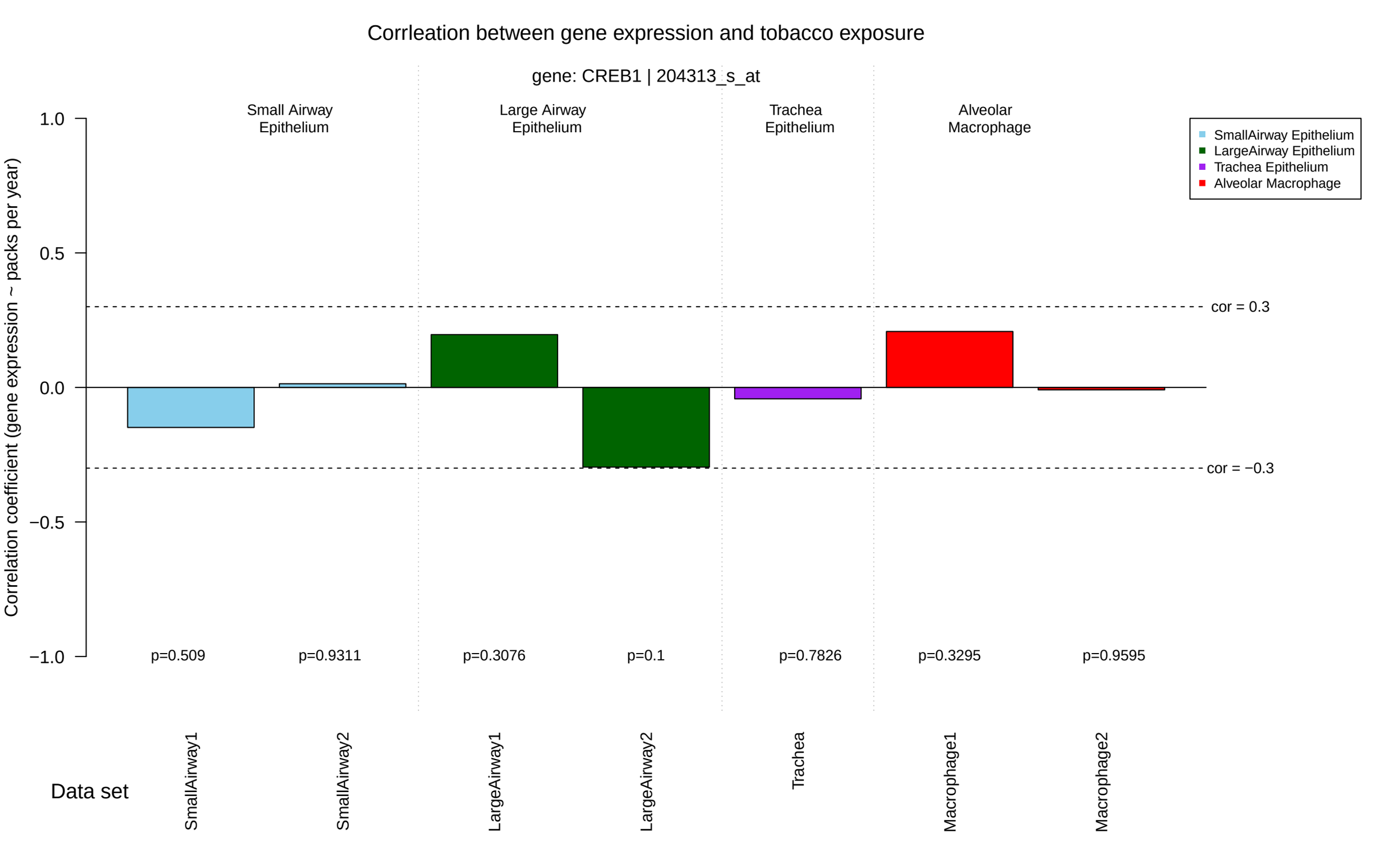

set: LargeAirway1 (p=0.3605)
CREB1 | 204313_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.052)
CREB1 | 204313_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2549)
CREB1 | 204313_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2532)
CREB1 | 204313_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4201)
CREB1 | 204313_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CREB1 | 204313_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.509
p=0.9311
p=0.3076
p=0.1
p=0.7826
p=0.3295
p=0.9595
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2